

Assignment 2

Name: _____

Student ID: _____

9 problems. Show all of your work.

Problem 1.The terms of a geometric series $a + ar + ar^2 + \cdots + ar^{n-1} + \cdots$ satisfy

$$a_5 + a_7 = 1500 \quad \text{and} \quad a_{11} + a_{13} = 187500,$$

where $a_n = ar^{n-1}$ denotes the n th term. Find $a_1 + a_2 + a_3$.**Problem 2.**

What is the sum of all multiples of 7 or 11 less than 1000?

Problem 3.What can we conclude by applying the n th term test in the following series?

A.
$$\sum_{n=1}^{\infty} \frac{(5n+3)^{2n}}{(7n^2+2)^n}$$

B.
$$\sum_{n=1}^{\infty} \frac{2 \ln(n)}{3n}$$

Problem 4.

Evaluate the following sums or show that they diverge.

A.
$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)}$$

B.
$$\sum_{n=0}^{\infty} e^{-2n}$$

C.
$$\sum_{n=0}^{\infty} e^{2n}$$

D.
$$\sum_{n=1}^{\infty} \left(1 + \frac{4}{n}\right)^n$$

E.
$$\sum_{n=2}^{\infty} \frac{2}{3^n}$$

Problem 5.

Answer questions A and B below for the following infinite series:

$$\sum_{k=1}^{\infty} \left(\frac{6}{k^2} - \frac{6}{(k+1)^2} \right).$$

- A. Does the n th-term test apply? Remember to fully justify your answer.
- B. Evaluate the series or show that it diverges.

Problem 6.

Express $8.818181818181818 \dots$ as a rational number, i.e. in the form $\frac{p}{q}$, where p and q are positive integers with no common factors.

Problem 7.

Find the sum of the convergent series $\sum_{n=0}^{\infty} \frac{(-25)^n}{(2n)!}$.

Problem 8.

Find the sum of the series $\sum_{n=0}^{\infty} \frac{1}{n!}$.

Problem 9.

Find the sum of the series

$$5 + 5^2 + \frac{5^3}{2!} + \frac{5^4}{3!} + \dots + \frac{5^{n+1}}{n!} + \dots$$